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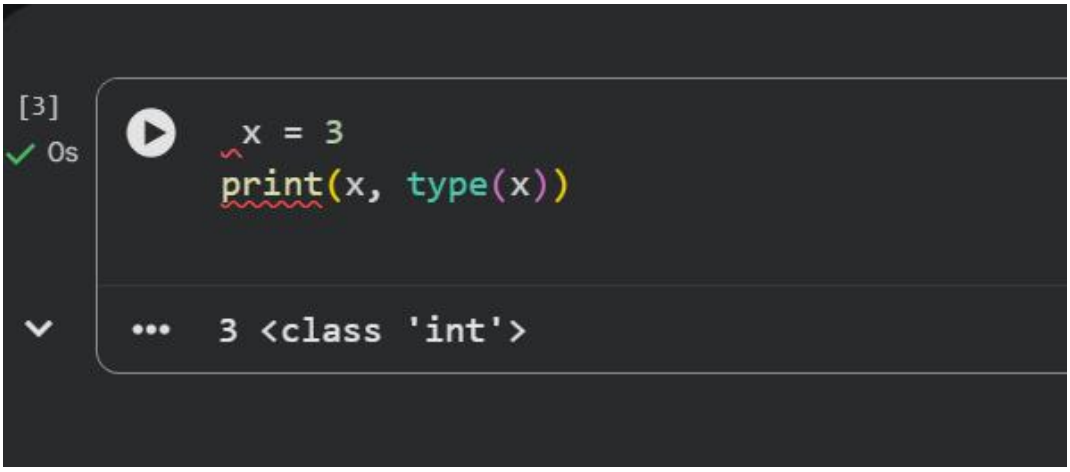
Course: Artifical Intelligence

Code Screenshots & Outputs

Numbers

x = 3

print(x, type(x))




```
x = 3
print(x + 1)  # Addition
print(x - 1)  # Subtraction
print(x * 2)  # Multiplication
print(x ** 2) # Exponentiation
```

```
x = 3
print(x + 1)  # Addition
print(x - 1)  # Subtraction
print(x * 2)  # Multiplication
print(x ** 2) # Exponentiation
```

```
4
2
6
9
```

X = 3

x += 1

print(x)

x *= 2

print(x)

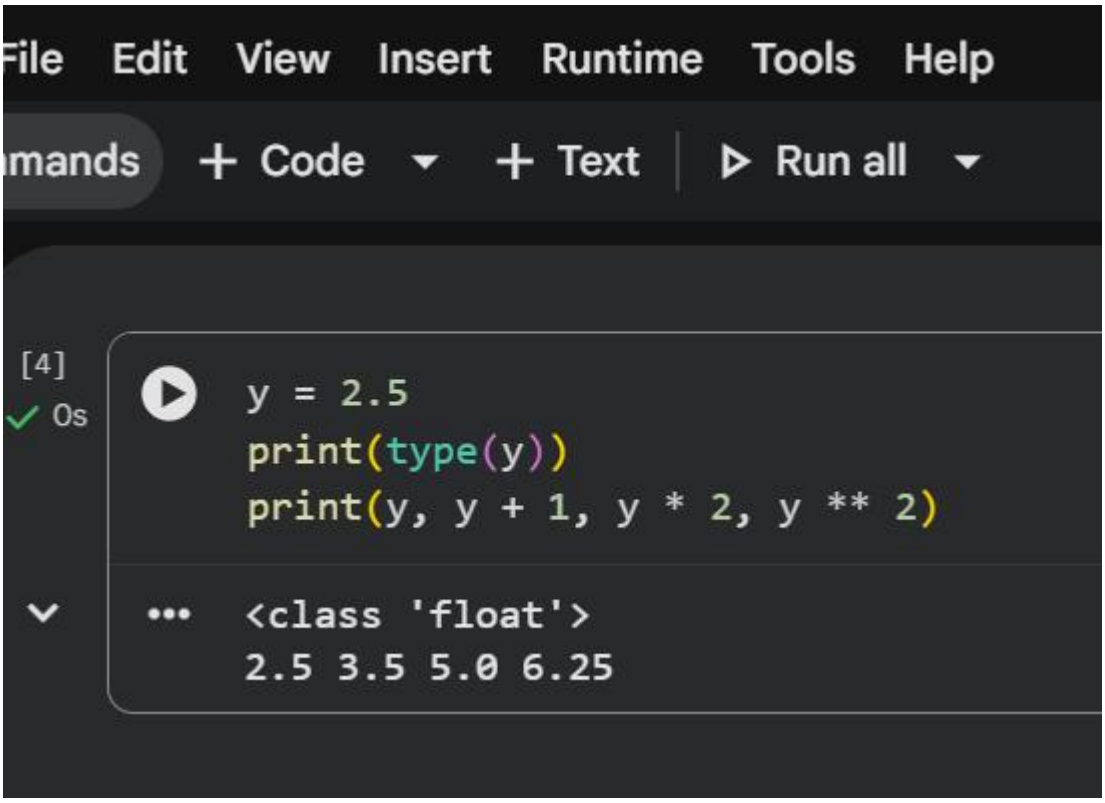
```
[3]
✓ 0s
x = 3
x += 1
print(x)
x *= 2
print(x)
```

```
4
8
```

y = 2.5

print(type(y))

print(y, y + 1, y * 2, y ** 2)



Booleans

t, f = True, False

print(type(t))

```
t, f = True, False
print(type(t))
```

```
<class 'bool'>
```

```
# Logical operations
print(t and f)  # Logical AND
print(t or f)   # Logical OR
print(not t)    # Logical NOT
print(t != f)   # Logical XOR
```

```
[6]
✓ 0s
# Logical operations
print(t and f)  # Logical AND
print(t or f)   # Logical OR
print(not t)    # Logical NOT
print(t != f)   # Logical XOR

False
True
False
True
```

Strings

```
# Strings
hello = 'hello'  # String literals can use single quotes
world = "world"  # or double quotes; it does not matter
print(hello, len(hello))
```

```
# Strings
hello = 'hello'  # String literals can use single quote
world = "world"  # or double quotes; it does not matter
print(hello, len(hello))
```

```
hello 5
```

String concatenation

hw = hello + ' ' + world

print(hw)

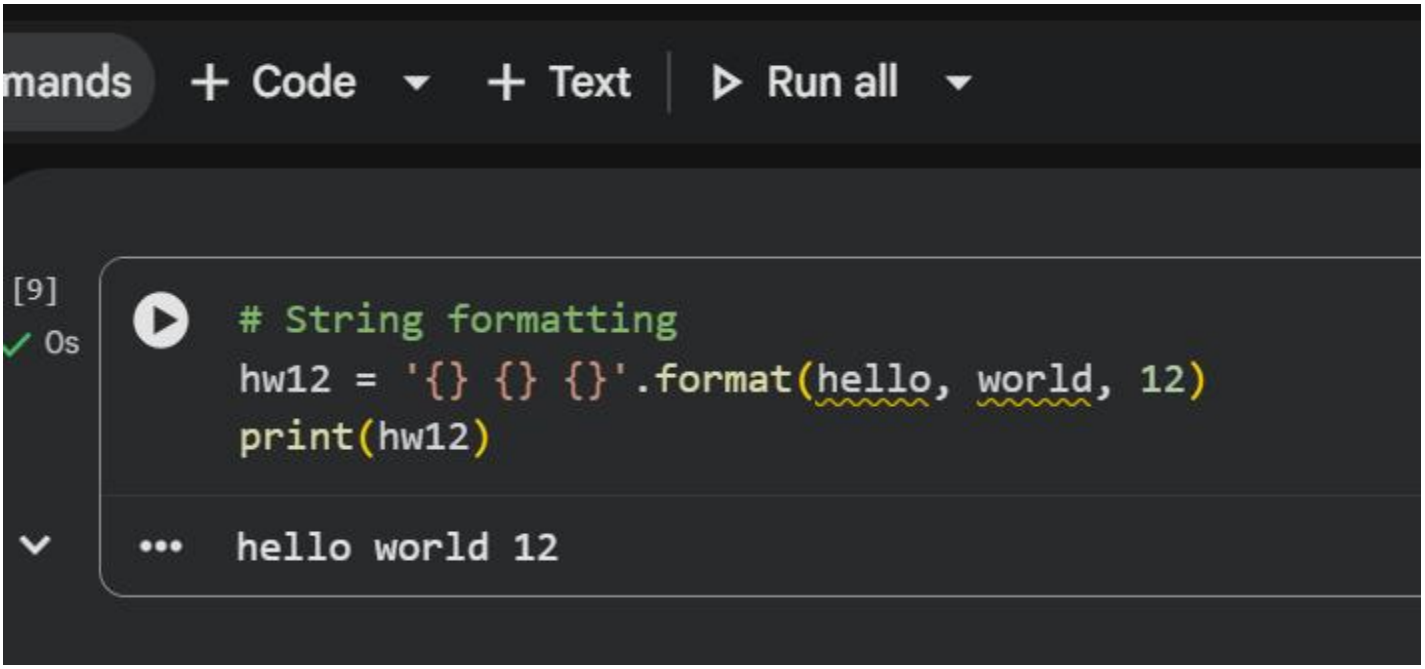
```
# String concatenation
hw = hello + ' ' + world
print(hw)
```

```
hello world
```

String formatting

hw12 = '{} {} {}'.format(hello, world, 12)

print(hw12)




```
# String methods
s = "hello"
print(s.capitalize())  # Capitalize a string
print(s.upper())       # Convert to uppercase
print(s.rjust(7))      # Right-justify with spaces
print(s.center(7))     # Center with spaces
print(s.replace('l', '(ell)'))  # Replace substrings
print('  world  '.strip())  # Strip whitespace
```

```python
# String methods
s = "hello"
print(s.capitalize())  # Capitalize a string
print(s.upper())       # Convert to uppercase
print(s.rjust(7))      # Right-justify with spaces
print(s.center(7))     # Center with spaces
print(s.replace('l', '(ell)'))  # Replace substrings
print('  world  '.strip())  # Strip whitespace
```

```
Hello
HELLO
  hello
 hello
he(ell)(ell)o
world
```

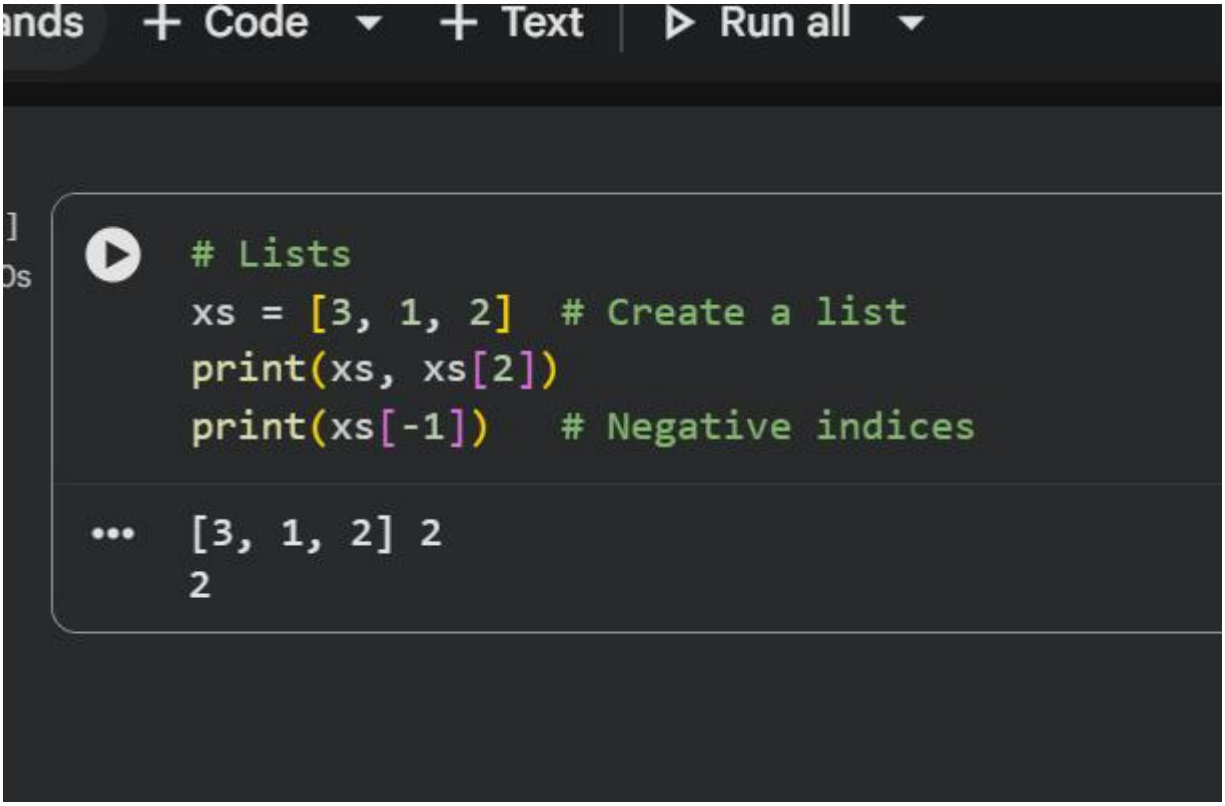
Lists

Lists

xs = [3, 1, 2] # Create a list

print(xs, xs[2])

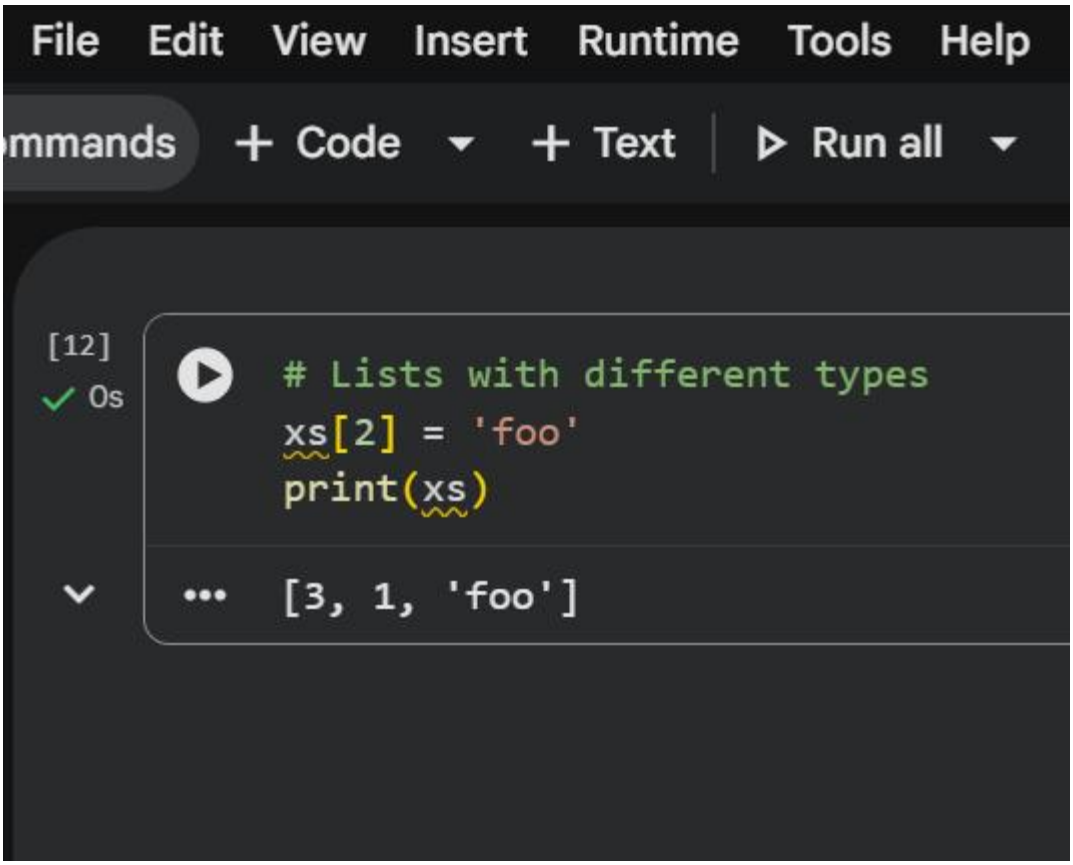
print(xs[-1]) # Negative indices



Lists with different types

xs[2] = 'foo'

print(xs)




```
# List methods
xs.append('bar')
print(xs)
x = xs.pop()
print(x, xs)
```

```
[13]
✓ 0s
# List methods
xs.append('bar')
print(xs)
x = xs.pop()
print(x, xs)

[3, 1, 'foo', 'bar']
bar [3, 1, 'foo']
```

Slicing

Slicing

nums = list(range(5))

print(nums)

```
ommands + Code + Text ▷ Run all
command palette (Ctrl+Shift+P)
[14]
✓ 0s
# Slicing
nums = list(range(5))
print(nums)

[0, 1, 2, 3, 4]
```

Various slice operations

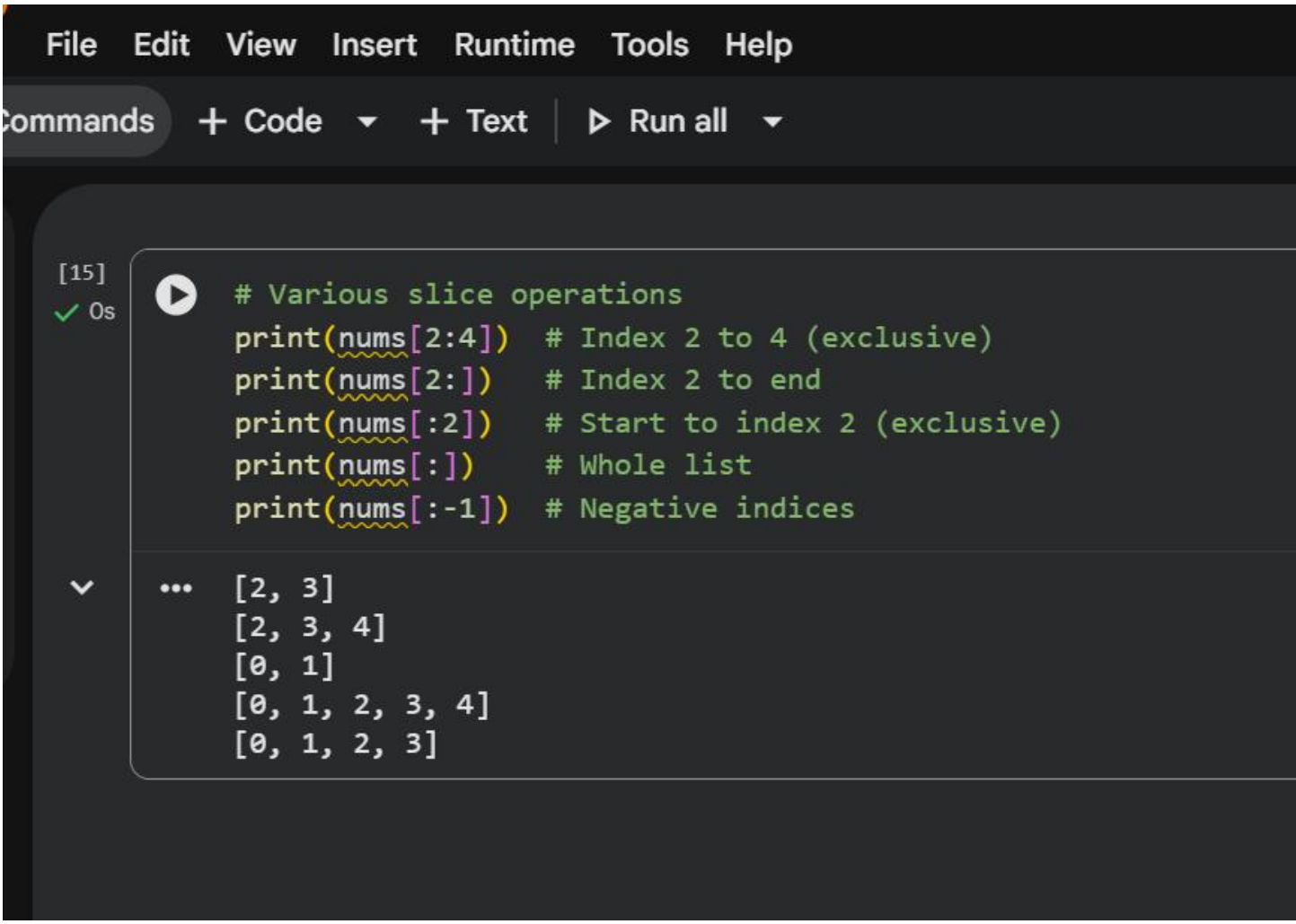
print(nums[2:4]) # Index 2 to 4 (exclusive)

print(nums[2:]) # Index 2 to end

print(nums[:2]) # Start to index 2 (exclusive)

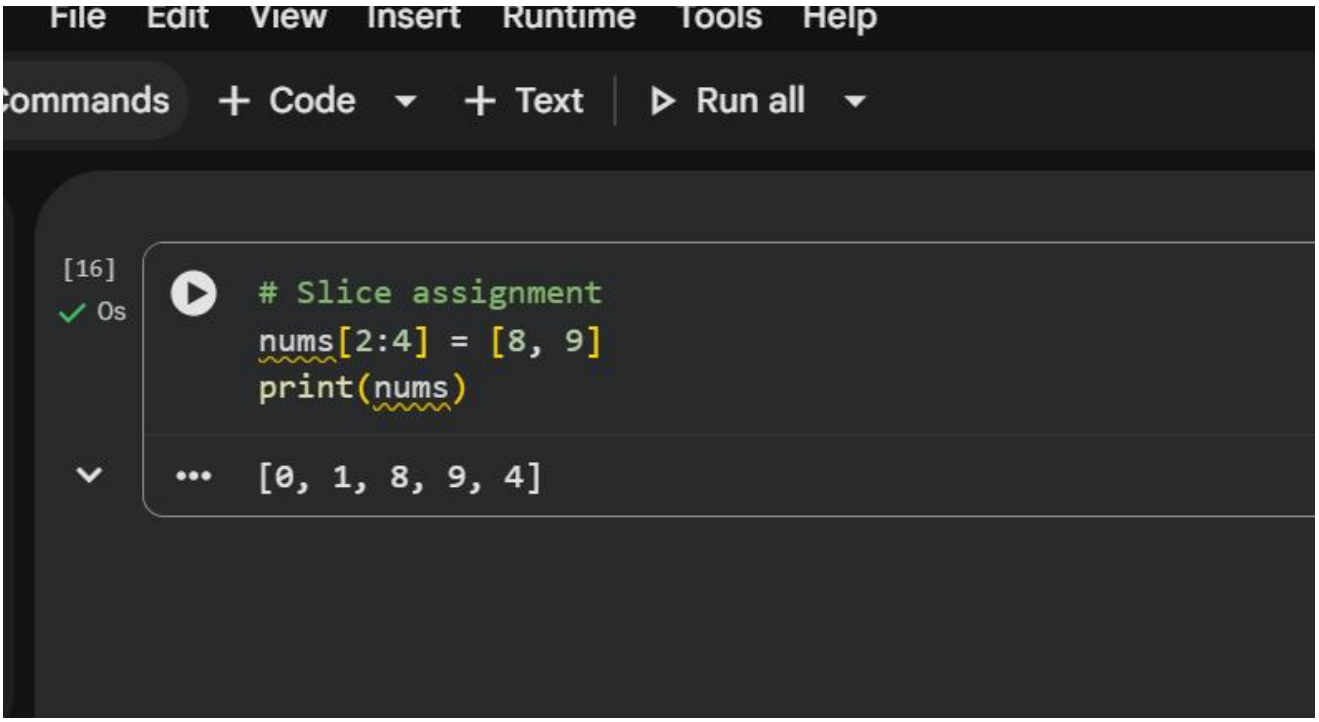
print(nums[:]) # Whole list

print(nums[:-1]) # Negative indices



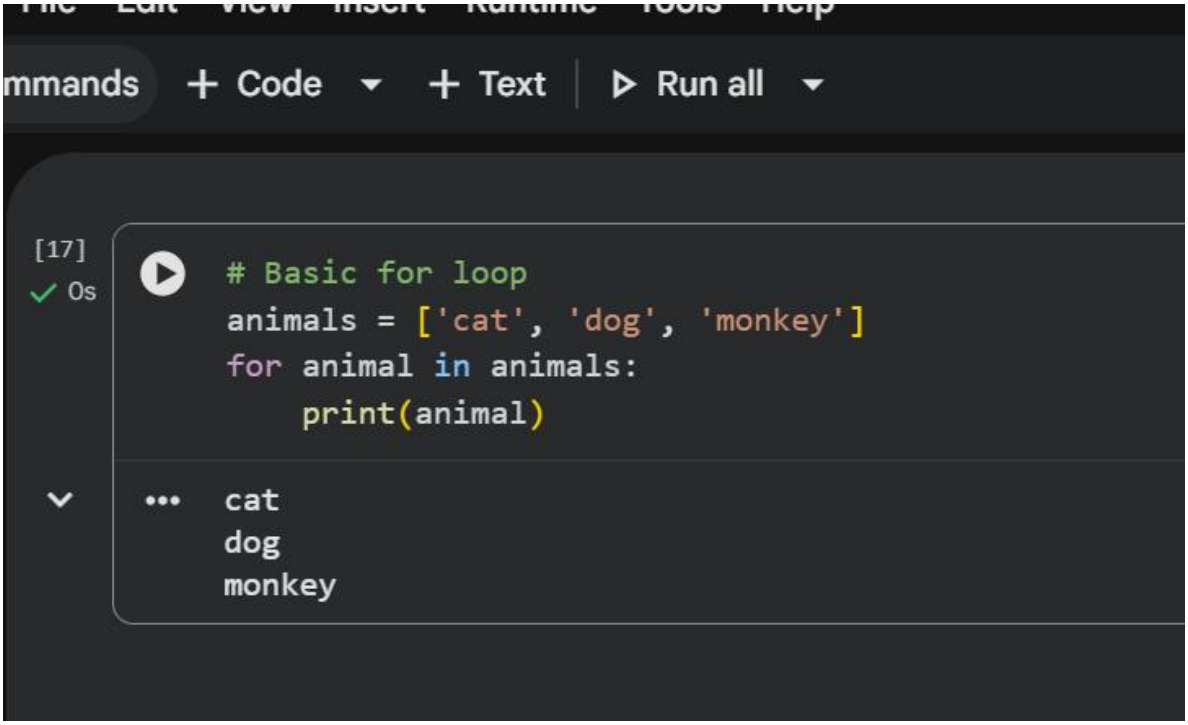
Slice assignment

```
nums[2:4] = [8, 9]
print(nums)
```



Loops

```
# Basic for loop
animals = ['cat', 'dog', 'monkey']
for animal in animals:
    print(animal)
```




```
# Range with step
for i in range(0, 5, 2):
    print("I am in iteration number:", i)
```

```
# Range with step
for i in range(0, 5, 2):
    print("I am in iteration number:", i)
```

```
I am in iteration number: 0
I am in iteration number: 2
I am in iteration number: 4
```

Strings

data = 'hello world'

print(data[0])

print(len(data))

print(data)

```
[2]: # Strings
data = 'hello world'
 print(data[0])
 print(len(data))
 print(data)

h
11
hello world

[ ]:
```

Numbers

data = 'hello world'

print(data[0])

print(len(data))

print(data)

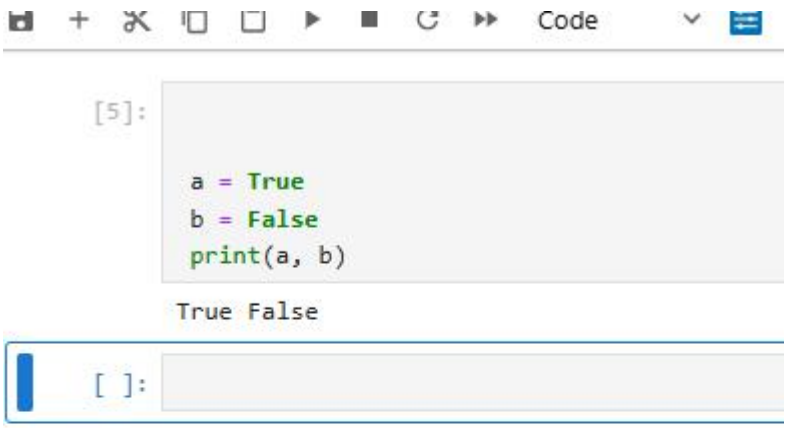
```
[4]: value = 123.1
     print(value)
     value = 10
     print(value)

     123.1
     10

[ ]:
```

Boolean

a = True

b = False

print(a, b)



Multiple Assignment

a, b, c = 1, 2, 3

print(a, b, c)

```
a, b, c = 1, 2, 3
print(a, b, c)
```

```
1 2 3
```

If-else Condition:

value = 99

if value >= 99:

print('That is fast')

elif value > 200:

print('That is too fast')

else:

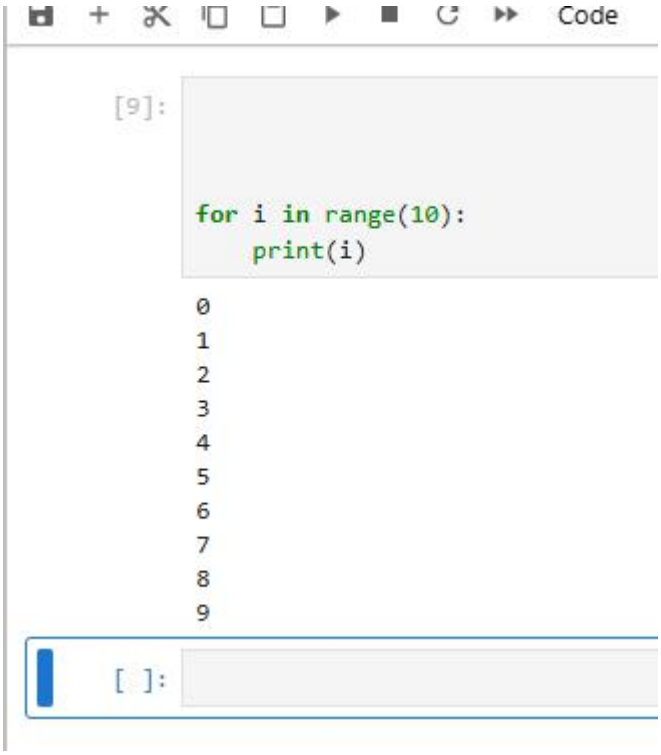
print('That that is safe')

```
value = 99
if value >= 99:
    print('That is fast')
elif value > 200:
    print('That is too fast')
else:
    print('That that is safe')
```

```
That is fast
```

For Loops:

```
for i in range(10):
    print(i)
```



While-Loop:

```
i = 0
while i < 10:
    print(i)
    i += 1
```

```
i = 0
while i < 10:
    print(i)
    i += 1
```

```
0
1
2
3
4
5
6
7
8
9
```

Tuple:

a = (1, 2, 3)

print(a)

```
a = (1, 2, 3)
print(a)
```

```
(1, 2, 3)
```

List:

mylist = [1, 2, 3]

print("Zeroth Value: %d" % mylist[0])

```
mylist.append(4)

print("List Length: %d" % len(mylist))

for value in mylist:

    print(value)
```

```
[12]:
mylist = [1, 2, 3]
print("Zeroth Value: %d" % mylist[0])
mylist.append(4)
print("List Length: %d" % len(mylist))
for value in mylist:
    print(value)

Zeroth Value: 1
List Length: 4
1
2
3
4
```

```
[ ]:
```

Dictionary:

```
mydict = {'a': 1, 'b': 2, 'c': 3}

print("A value: %d" % mydict['a'])

mydict['a'] = 11

print("A value: %d" % mydict['a'])

print("Keys: %s" % mydict.keys())

print("Values: %s" % mydict.values())

for key in mydict.keys():

    print(mydict[key])
```

[13]:

```
mydict = {'a': 1, 'b': 2, 'c': 3}
print("A value: %d" % mydict['a'])
mydict['a'] = 11
print("A value: %d" % mydict['a'])
print("Keys: %s" % mydict.keys())
print("Values: %s" % mydict.values())
for key in mydict.keys():
    print(mydict[key])
```

```
A value: 1
A value: 11
Keys: dict_keys(['a', 'b', 'c'])
Values: dict_values([11, 2, 3])
11
2
3
```

[ ]:

Sum function

def mysum(x, y):

return x + y

[14]:

```
def mysum(x, y):
    return x + y
```

[ ]:

[ ]:

Test sum function

print(mysum(1, 3))

[15]:

```
print(mysum(1, 3))
```

```
4
```